Isotopes and Average Atomic Mass

Answer key with worked solutions

A. Understanding

1. What the atomic mass represents.

> The **weighted average** mass of the element's naturally occurring isotopes, based on their abundances.

2. Which isotope is more abundant?

> The **20.0 u** isotope: the average (20.2 u) is much closer to 20.0 u than to 22.0 u, so the lighter isotope must be far more common.

B. Calculating average atomic mass

Each abundance is written as a fraction (percentage ÷ 100), which has no unit.

3. Boron.

> (0.199 × 10.013 u) + (0.801 × 11.009 u) = **10.81 u**

4. Rubidium.

> (0.7217 × 84.912 u) + (0.2783 × 86.909 u) = **85.47 u**

5. Silver.

> (0.5184 × 106.905 u) + (0.4816 × 108.905 u) = **107.87 u**

6. Silicon.

> (0.9223 × 27.977 u) + (0.0468 × 28.976 u) + (0.0309 × 29.974 u) = **28.09 u**

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C. Challenge

7. Abundances of copper isotopes.

> Let x = fraction of copper-63 (no unit): $(62.930\ \text{u})x + (64.928\ \text{u})(1 - x) = 63.546\ \text{u}$
>
> $64.928\ \text{u} - (1.998\ \text{u})x = 63.546\ \text{u}$, so $x = \dfrac{1.382\ \text{u}}{1.998\ \text{u}} = 0.692$
>
> **Copper-63: 69.2%, copper-65: 30.8%**

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